

Guided Quiz: Understanding OLS

Follow the logic of OLS and explain each idea in your own words.

Name: _____

Student ID: _____

1. What are we trying to learn?

Consider the population model

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{u}.$$

What does $\boldsymbol{\beta}$ represent?

What does $\mathbf{u}$ contain?

2. How is the OLS estimator defined?

Explain the difference between the following objects.

- $\boldsymbol{\beta}$: _____
- $\mathbf{b}$: _____
- $\hat{\boldsymbol{\beta}}$: _____

What does OLS try to minimize?

In words, how does minimizing this quantity determine the regression line?

3. What are OLS residuals?

What is an OLS residual? Complete the expression and explain it in words.

$$\hat{u}_i = \underline{\hspace{2cm}}$$

Why can some residuals be positive and others negative?

4. A mechanical property of OLS

OLS residuals satisfy $\mathbf{X}'\hat{\mathbf{u}} = \mathbf{0}$. What does this mean intuitively?

Is this a population assumption or a property that follows automatically from fitting OLS?

Population assumption

Mechanical property of OLS

5. Why is $\hat{\beta}$ different from β ?

Why do different random samples generally produce different values of $\hat{\beta}$?

What does the following expression tell us?

$$\hat{\beta} - \beta = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{u}.$$

6. What is exogeneity?

The exogeneity assumption is

$$E(\mathbf{u} \mid \mathbf{X}) = \mathbf{0}.$$

Explain this assumption in plain language.

Why might this assumption fail? Give one example.

7. Sample residuals versus population errors

Explain the difference between

$$\mathbf{X}'\hat{\mathbf{u}} = \mathbf{0} \quad \text{and} \quad E(\mathbf{u} \mid \mathbf{X}) = \mathbf{0}.$$

Which statement is automatically satisfied by OLS? _____

Which statement must be justified using economic reasoning or research design?

8. What does unbiasedness mean?

Under exogeneity,

$$E(\hat{\beta} | \mathbf{X}) = \beta.$$

Explain this result in words.

Does unbiasedness mean that $\hat{\beta} = \beta$ in every sample?

☐ Yes☐ No

Explain briefly.

9. What determines the precision of OLS?

What is the difference between an estimate and its standard error?

What does a small standard error tell us?

Name two factors that generally make an OLS estimate more precise.

1. _____

2. _____

10. Why do we estimate the error variance?

Why is σ^2 unknown?

Why do we use residuals $\hat{u}_i$ to learn about unobserved errors u_i ?

Why do we divide SSR by $n - p$, rather than by n ?

11. Why does the normal distribution appear?

As the sample size becomes large, the standardized OLS estimator is approximately normally distributed. Why is this useful?

What can we construct using this approximate sampling distribution?

12. Statistical significance

What question does a hypothesis test answer?

What does a confidence interval tell us?

If an estimate is statistically significant, is it necessarily economically important?

☐ Yes☐ No

Explain.

13. Statistical inference versus causal inference

Complete the comparison.

Statistical inference asks: How _____?

Causal inference asks: What would happen to _____?

Can a biased estimate have a very small standard error? ☐ Yes ☐ No

Can a relationship be statistically significant but not causal? ☐ Yes ☐ No

Explain why.

14. What does Gauss–Markov tell us?

OLS is BLUE. What does “unbiased” mean here?

What does “best” mean here?

Does Gauss–Markov establish that the OLS estimate is causal? ☐ Yes ☐ No

Does it say that OLS is best among every imaginable estimator? ☐ Yes ☐ No

Among which class of estimators is OLS best?

Final summary

Complete the logical sequence using: *OLS*, *sampling uncertainty*, *exogeneity*, *unbiasedness*, *statistical inference*.

1. _____ chooses the coefficients that minimize squared residuals.
2. Because samples differ, the OLS estimate is subject to _____.
3. The _____ assumption connects the regressors to the unobserved error.
4. Under this assumption, OLS has the property of _____.
5. Standard errors and approximate normality allow us to conduct _____.